

Computación en Internet I

Semana 4

Nicolás J. Salazar E.

TCP

UDP

Multiplexing and Demultiplexing

SSL / TLS

QUIC

TCP - Transmission Control Protocol

Three-way Handshake

PC A $\rightarrow$ TCP SYN $\rightarrow$ PC B

PC A $\leftarrow$ TCP SYNC/ACK $\leftarrow$ PC B

PC A $\rightarrow$ TCP ACK $\rightarrow$ PC B

TCP

SRC. PORT	DST. PORT
SEQUENCE NUMBER	
ACK NUMBER	
CHECKSUM	URGENT POINTER
DATA	

TCP flags

- URG - URGENT
- ACK – ACKNOWLEDGE
- PSH – PUSH
- RST – RESET
- SYN – SYNCHRONIZE
- FIN – FINISH

UDP - User Datagram Protocol

Conocido por ser *Best-effort delivery*

UDP

SRC. PORT	DST. PORT
LENGTH	CHECKSUM
DATA	

Multiplex and Demultiplex

IANA

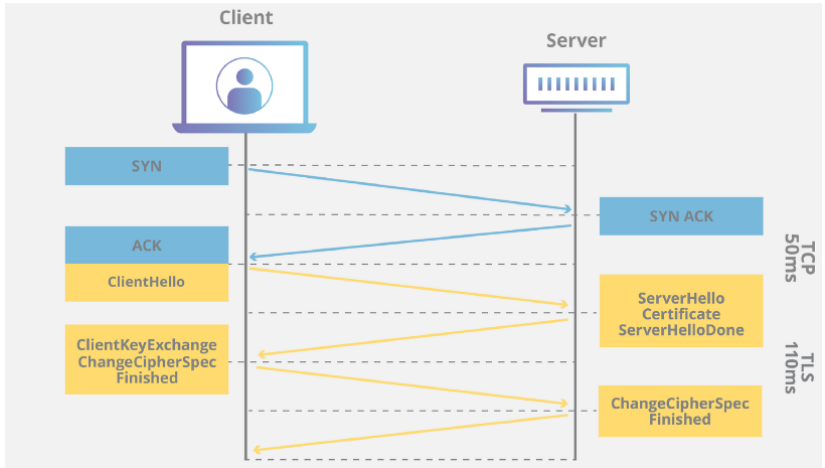
- Puertos Bien conocidos [0 - 1023]
- Puertos de uso privado [1024 - 49151]
- Puertos Efimeros [49152 - 65535]

SSL / TLS

- SSL - Secure Socket Layer (3.0 - 1996)
- TLS - Transport Layer Security
- Encriptación
- Autenticación
- Integridad

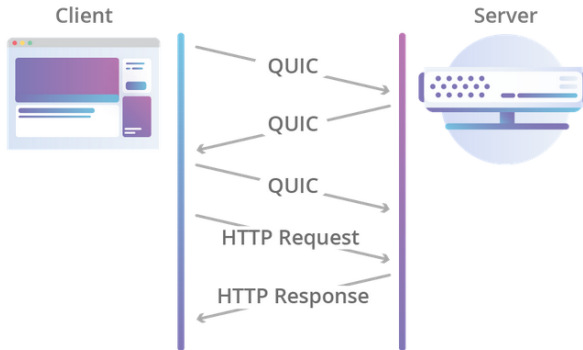
TLS handshake

- Client Hello
- Server Hello
- Authentication
- Premaster Secret
- Private Key
- Session Key
- Client Ready
- Server Ready
- Secure Symmetric Encryption Achieved

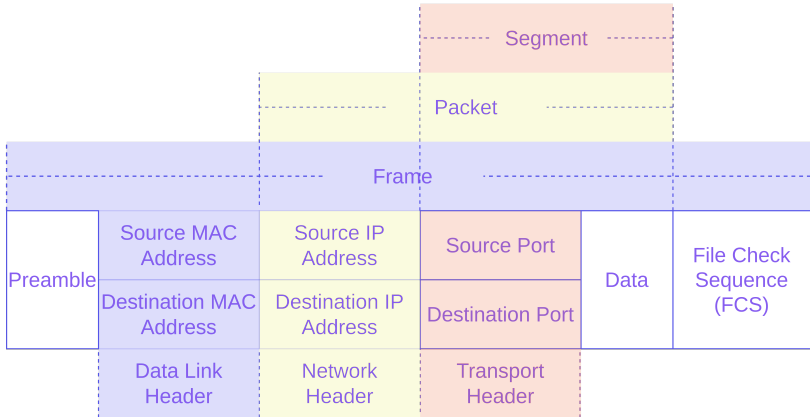


QUIC - Quick UDP Internet Connections

HTTP Request Over QUIC



Segment





Gracias